



Knowledge Representation – OSP-5

Definition

Knowledge Representation (KR) is a branch of artificial intelligence that focuses on how to **formally structure knowledge** about the world so that a computer system can **understand, reason, and act** upon it.

It involves encoding information in a format that allows machines to **simulate human-like understanding** and **decision-making**.

Objectives

- To represent real-world knowledge in a form that a computer can utilize.
 - To support **inference, decision-making, and problem-solving**.
 - To make knowledge **accessible, comprehensible, and manipulatable** by machines.
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Characteristics of Good Knowledge Representation

1. **Representational Adequacy**
 - Ability to represent all kinds of knowledge required for the task.
2. **Inferential Adequacy**
 - Ability to derive new knowledge using logical inference.



3. Inferential Efficiency

- Ability to infer efficiently and practically within time constraints.

4. Acquisitional Efficiency

- Ability to acquire new knowledge easily and update existing knowledge.

Types of Knowledge Represented

Type	Example
Factual Knowledge	"Paris is the capital of France."
Procedural Knowledge	"How to solve a quadratic equation."
Heuristic Knowledge	"If someone is coughing, they might be sick."
Meta-Knowledge	"Knowing what you know and don't know."

Approaches to Knowledge Representation

1. Logic-Based Representation

- Uses formal logic (e.g., propositional or predicate logic).



- Enables **rigorous inference** and **truth validation**.
- Example: $\forall x \text{ (Human}(x) \rightarrow \text{Mortal}(x))$

2. Semantic Networks

- Graph-based structure where **nodes** represent entities and **edges** represent relationships.
- Example: $\text{Dog} \rightarrow \text{is-a} \rightarrow \text{Animal}$

3. Frames

- Structured data objects representing **stereotyped situations** or objects.
- Each frame includes slots (attributes) and values.
- Useful for representing **common-sense knowledge**.

4. Production Rules

- Rule-based format: **IF condition THEN action/conclusion**.
- Common in expert systems.
- Example: $\text{IF fever AND cough THEN suspect flu}$

5. Ontologies

- Formal representation of a **set of concepts** and relationships in a domain.
- Often used in the **semantic web** and **knowledge graphs**.



Importance in AI

- Essential for **machine reasoning**, **expert systems**, **natural language understanding**, and **planning**.
 - Bridges the gap between **raw data** and **intelligent behavior**.
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Example

Suppose a robot needs to serve tea:

- **Factual Knowledge**: Tea is in the kitchen.
 - **Procedural Knowledge**: Steps to make tea.
 - **Heuristic Knowledge**: Tea is often served hot.
 - **Meta-Knowledge**: The robot knows it doesn't know where the sugar is.
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Summary Table

Feature	Description
Purpose	Enable machines to process and reason with knowledge
Forms	Logic, semantic networks, frames, rules, ontologies



Applications Expert systems, robotics, natural language processing

Benefits Enhances reasoning, decision-making, and learning